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Fall-risk assessment of aged workers using wearable inertial measurement units based on machine learning

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ABSTRACT

Falls are the major cause of fatal and nonfatal incidents among aged workers. Estimating individual fall risk is crucial for occupational protection and public health. This study aimed to develop a fall-risk assessment method based on Machine Learning (ML) and Inertial Measurement Units (IMUs). A total of 28 aged workers (60 to 80 years old) participated in this study, recruited from community sanitation workers and janitors. They were categorized into high and low fall-risk groups based on functional gait assessment. Each participant performed 5 sets of motion experiments in the lab, covering 6 daily motions. We gathered their kinematic data with IMUs for training ML-based fall-risk assessment models. Five classic ML classifiers were optimized using grid search, and a Convolutional Neural Network and Long Short-Term Memory (CNN-LSTM) hybrid deep neural network was built to recognize walking motion from kinematic data for input into the fall-risk assessment model. The results demonstrated that ML classifiers trained with walking motion data exhibited superior performance. Cross-validation revealed that the optimized Random Forest classifier, with IMU on the right upper leg, reached 91.3 % accuracy. Additionally, the CNN-LSTM models were evaluated using F1-score, which can balance accuracy and coverage of model performance evaluation. The CNN-LSTM achieved a maximum F1-score of 95.18 %, proving their effectiveness in extracting walking motion data. These findings indicate that the developed IMU-based ML method exhibits excellent performance in assessing fall risk, potentially allowing aged workers to assess their fall risk before clinical diagnosis by embedding it in wearable IMUs.

1. Introduction

Population aging has emerged as a pressing global concern, precipitating an elongation of working lives and a surge in the proportion of aged workers over age 50 (Farrow & Reynolds, 2012; Peng & Chan, 2020) across diverse industries encompassing mental and manual labor practitioners. This demographic shift presents profound challenges in ensuring occupational safety. Certain regions and industries still have a significant proportion of aged workers aged 50 and even 60 and above (Schwatka et al., 2013; Farrow & Reynolds, 2012), coupled with current economic and policy trends, this proportion is still increasing (Varianou-Mikellidou et al., 2021). The number of aged workers (over 55 years) is expected to grow by 28 % between 2020 and 2030 in emerging countries, accounting for 18 % of the total labor force (Harasty & Ostermeier, 2020). This escalating proportion of aged workers has led to emerging safety concerns and amplified difficulties in ensuring occupational protection. While aging is often accompanied by a decline in physical function and cognitive abilities among workers (Bargiotas et al., 2023; Kooij et al., 2020). Recent research suggests that the wealth of work experience and knowledge possessed by aged workers partially offsets the occupational risks associated with reduced physical function (Bravo et al., 2022). But the experience might not be enough to diminish the accidents (Halabi et al., 2022). Nonetheless, it is undeniable that aged workers face higher injury costs (Schwatka et al., 2012) and endure more severe consequences following an injury compared to their younger counterparts (Pilipiec et al., 2021). Given the unique occupational capabilities and increased injury costs among aged workers, conducting specific research on their safety is both necessary and valuable.

Falls constitute the most prevalent fatal or nonfatal injuries among a range of occupational injuries associated with aging (Sherrington et al., 2020). According to the World Health Organization, an estimated

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684,000 individuals globally succumb to falls each year. During the last three decades, mortality rates due to falls increased (GBD, 2022). The incidence of falls is significantly impacted by specific working environments and tasks (Ahmad et al, 2017). Particularly in manual labor, numerous tasks pose challenges to maintaining stable balance control (Antwi-Afari et al., 2017). For instance, the majority of sanitation workers are comprised of individuals aged between 50 and 80 years, frequently face the challenging task of working on slippery roads in adverse weather conditions. Aged construction workers face the risk of tripping over intricate support structures (Dong et al., 2012). The likelihood and severity of falls escalate with advancing age (Moreland et al., 2020), as aged workers experience muscle mass loss, diminished strength (Rezuş et al., 2020), and degeneration of vestibular function. These factors further exacerbate their susceptibility to falls and increases the severity of resulting consequences. Aged workers experienced fall injuries more often than younger workers (Stoesz et al., 2020; Bravo et al., 2022), and 20 to 30 % of them have moderate-to-severe injuries (Bhasin et al., 2020). Fatal falls are higher among older workers compared to younger workers (Haagsma et al., 2020). Fall-related injuries also impose a significant financial burden on families and society, while simultaneously consuming substantial public medical resources. Costing approximately 50 billion US dollars annually in the United States (Burns et al., 2016) and 25 billion euros in the European Union (Hartholt et al., 2011). Therefore, conducting fall risk assessment and prevention studies among aged workers is crucial for safeguarding this vulnerable demographic, promoting societal advancement, and enhancing individual and overall societal safety levels.

Assessing fall risk serves as the foundation for various studies related to falls, such as investigating the impact of different factors on falls and evaluating the effectiveness of fall prevention strategies. The measures used to assess fall risk can be categorized into two types: subjective and objective measures (Guo et al., 2020). The risk assessment based on perception, which employs clinical assessment scales, is a commonly applied subjective method primarily executed by experts in geriatrics, nursing practitioners, and physiotherapists (Perell et al., 2001; Montero-Odasso et al., 2022). Commonly used clinical assessment scales include Balance Evaluation Systems Test (BESTest) (Horak et al., 2009), Dynamic Gait Index (DGI) (Jonsdottir & Cattaneo, 2007), Functional Gait Assessment (FGA) (Wrisley et al., 2004), Berg Balance Scale (BBS), the Timed Up and Go Test (TUG) (Blum & Korner-Bitensky, 2008), etc. Most of the methods have demonstrate efficacy in evaluating the fall risk of specific groups (mainly the elderly, Parkinson's patients, stroke patients, etc.). Primarily dependent on subjective judgment, these assessment methods are prone to biases that may be affected by the evaluator's personal traits or prior experiences (Morse, 2006). In contrast, objective assessment techniques utilize equipment to ascertain the level of fall risk, offering a more unbiased assessment results compared to subjective approaches. Center of pressure (COP) trajectory-based measures represent the conventional and most prevalent objective measures (Quijoux et al., 2020; Qiu & Xiong, 2015), occasionally serving as the golden standard. Nevertheless, acquiring COP data necessitates the use of force plates (e.g., AMTI Force Plate, AMTI, Watertown, MA, USA), which are expensive, challenging to set up, and cumbersome to transport (Richmond et al., 2021; Hua et al., 2018). With advancements in data transmission methods and integrated techniques, the utilization of Inertial Measurement Units (IMUs) for human motion capture and posture identification has grown considerably (Li et al., 2022; Subramaniam et al., 2022). Owing to its compact dimensions, IMUs can be comfortably attached to the human body, enabling the collection of movement data (Bruijn et al., 2010). Its low cost makes it easily accessible to the public (Bai et al., 2020). IMU offers the advantage of no spatial restrictions in signal detection, and its feasibility and reliability for human motion research in behavior detection (Ossig et al., 2016) and movement quality (Dixon et al., 2018) have been verified in previous studies. Given that human fall risk assessment measures depend on changes and characteristics in human behavior and posture (Pfister et al., 2014), assessing fall risk based on IMU is feasible and it has been proven to be a potentially useful screening tool for balance impairments and fall risk (McManus et al., 2022).

In IMU-based fall-related studies, postural stability and gait stability are the most frequently investigated aspects (Patel et al., 2020; Ghislieri et al., 2019). Regarding gait stability assessment, Jebelli et al. (2016) confirmed the applicability of IMU-based measures for assessing the gait stability of iron workers. Some scholars have also employed IMUs for fall risk assessment. Atrsaei et al. (2022) used an IMU to extract potential information present in subjects during tests. They discovered that the additional information supplied by IMU enabled more accurate detection of subjects with a high fall risk. Nonetheless, the quantity of these studies continues to be limited. A primary limitation of IMU-based approaches is the necessity for multiple areas of expertise and human intervention in processing raw data and extracting meaningful features for analysis (Tosi et al., 2018). As a result, IMU raw data cannot be directly utilized for fall risk assessment.

Currently, fall-risk assessment predominantly relies on functional assessment instruments conducted by physical therapists. If the difficulty of complex IMU data processing can be solved, IMU-based measures could be used for fall risk assessment. This would offer more convenient, affordable, and objective methods for aged workers to assess fall risk. Addressing the aforementioned issue, Machine Learning (ML) might serve as a potent solution, which has exceptional benefits in data analysis and classification. In pertinent studies, ML is primarily utilized for analyzing human posture and movement data captured by IMUs. ML methods can extract deep-level data features and offer superior benefits over threshold-based methods (Chakraborty & Sorwar, 2022) or tackling supervised classification and unsupervised clustering. Recently, investigators have started applying ML into the domain of fall research concerning older adults. In recent years, researchers utilized ML techniques to develop a fall detection algorithm (Doulamis & Doulamis, 2018) and fall-risk detection method (Dubois et al., 2019). Some researchers are already focusing on IMU-based fall prediction. Yu et al. (2020) developed a novel hybrid DNN to predict pre-impact falls for the elderly, achieving over 90 % sensitivity. These studies showcase the capabilities of IMU-based measures in analyzing deep and extra features of motion data, indicating their significant potential for assessing fall risks. The review study of Montesinos et al. (2018) stated that, in the research of IMU-based fall risk assessment of older adults, it is crucial to identify the optimal combination of three factors: where to place the IMUs, which motion task to be performed and which features should be extracted and analyzed, in order to develop effective assessment methods.

In summary, the escalating issue of global population aging and the corresponding increase in aged workers has heightened the severity of risks and consequences associated with falls. Consequently, there is an urgent need to conduct research on fall risk assessments and interventions specifically tailored for aged workers. This research aims to develop a fall risk assessment method for aged workers, utilizing both IMU and ML technologies. It also seeks to explore the impact of motion types, feature selections and IMU configurations in developing fall-risk assessment models based on IMUs to identify the optimal combination, address occupational safety challenges, and overcome existing research limitations. This method extracts motion data from the daily behavior of aged workers and accurately assesses their fall risk. It should be noted that this paper focuses on evaluating intrinsic fall risk for an individual, since it's reasonable and marketable to evaluate a worker's inherent level of fall risk.

2. Methods

2.1. Experimental design and data collection

2.1.1. Participants

Twenty-eight aged workers (50–80 years old) participated in this

experiment (twenty-three males and five females; age 67.18 ± 5.91 years; height 164.79 ± 7.15 cm; body mass 65.03 ± 15.52 kg). They were recruited from community sanitation workers and janitors, professions that typically employ aged workers and involve physically demanding tasks. The recruitment of participants was based on their willingness and ability to complete the experiment. The inclusion criteria exclude individuals with some diseases that could affect experiment completion or significantly affect physical balance, such as Parkinson's disease. All participants signed informed consent after being informed about the details of the experiment and were accompanied throughout the experiment by a caregiver to prevent any potential injuries or accidents during testing sessions. The study protocols received approval from the Ethics Review Board at China University of Geosciences.

2.1.2. Experimental apparatus

(1) Functional gait assessment (FGA)

In this study, FGA was used to pre-assess the fall risk of participants and categorize them into high fall-risk group (FGA score ≤ 22) and low fall-risk group (FGA score > 22), cutoff scores refer to the previous study (Klima & Hood, 2020), which informed data labeling for the ML classifier. FGA was designed to assess fall risk and gait stability among older adults (Wrisley et al., 2004). It has demonstrated promising sensitivity and specificity in classifying fall risk (Marques et al., 2021). Therefore, FGA can serve as a reliable reference standard. Fig. 1 shows the items of FGA and rules for subject grouping. The details of the FGA test procedure can be found in research of Wrisley et al. (2004).

(2) IMU sensors and configuration

For IMU data collection, the Xsens MTw Awinda (Xsens Inc., Enschede, The Netherlands) motion capture system was used to record acceleration data at a frequency of 100 Hz. Six IMUs were placed on the following body parts: sternum (T8, eighth thoracic vertebra), pelvis, right forearm, right upper leg, right knee, and right ankle, as shown in Fig. 2(a). To eliminate the potential impact of irrelevant variables on dynamic stability, the subjects were provided with rubber-soled shoes of the same model in sizes ranging from EU 36 to EU 44. All participants wore the provided shoes of their preferred size according to their feet size. After the experiment, the shoes underwent disinfection using alcohol. Three-dimensional acceleration and orientation data were collected. The three-dimensional acceleration comprises anteroposterior direction (DIR_AP), mediolateral direction (DIR_ML), and inferior-superior direction (DIR_IS). IMU-based data in the three-dimensional composite direction (DIR_3D) were utilized to develop fall-risk assessment models in this study, owing to their efficiency (Lee & Park, 2020). Fig. 2(b) shows the diagram of IMU directions, DIR_3D data were computed using the following equations.

$$\overrightarrow{DIR_2D} = \sqrt{(\overrightarrow{DIR_AP})^2 + (\overrightarrow{DIR_ML})^2} \quad (1)$$

$$\overrightarrow{DIR_3D} = \sqrt{(\overrightarrow{DIR_2D})^2 + (\overrightarrow{DIR_IS})^2} \quad (2)$$

2.1.3. Experimental task

The entire experiment comprised two tasks, fall risk assessment using the FGA (Task A) and motion data collection using IMU (Task B).

(1) Task A: fall risk assessment using FGA

Task A was carried out according to the process shown in Fig. 1. The subjects completed the ten evaluation items of the FGA test (see Fig. 1) in order. Throughout testing, a professional experimenter guided the subjects in completing the ten assessment tasks and scored them. Subjects were categorized into high fall-risk group or low fall-risk group based on their scores.

(2) Task B: motion data collection using IMU

Each participant was instructed to perform six motions (see Fig. 3), with IMU data being collected during the process. Every motion was performed five times in a randomized order (Guo & Xiong, 2017) by each participant. Prior to each task, subjects were required to rest for a sufficient duration (5–15 min, contingent upon the subject's sense of complete relaxation). The IMU data collection process referred to previous research of Yang et al. (2019). At the beginning of the experiment, participants were instructed to attach IMUs and execute specific motions. During this process, IMU data was collected with the aid of the matching software, Xsens MT Manager 4.6.

2.2. Fall-risk assessment based on machine learning and IMU

2.2.1. Training data set creation and data labeling

Initially, the data underwent filtering via a fourth-order zero-phase low-pass Butterworth filter (Yu et al., 1999) to minimize the impact of data noise and interference. The filtered IMU data, encompassing synthetic acceleration in the DIR_3D direction for six motions from 28 participants, served as the training dataset. According to the FGA score (Wrisley & Kumar, 2010), participants were categorized into high fall-risk group (positive samples) and low fall-risk group (negative samples). Table 1 shows the rules of subject grouping and data labeling.

Following this, it is imperative to consider the matter of sample imbalance within the dataset, as this will negatively impact classifier performance (Thabtah et al., 2020). This paper used the undersampling method (Mohammed et al., 2020) to address this problem. Suppose there are n sets of positive samples and m sets of negative samples. If $m \neq n$, undersampling is applied to process the training samples. If $n < m$, randomly select n sets of samples from the negative samples, along with

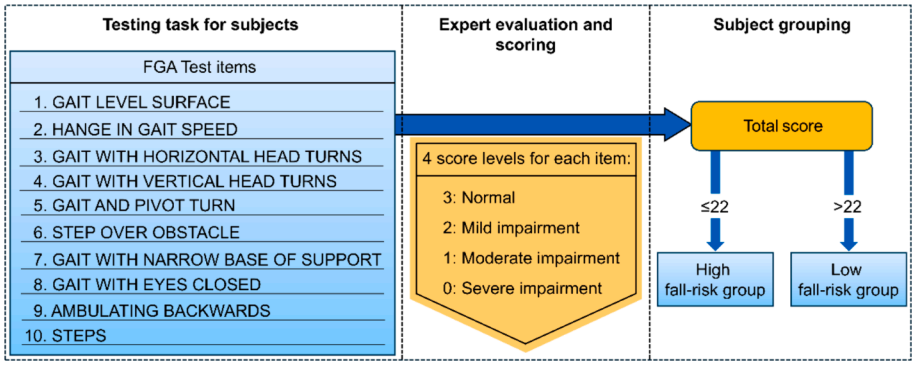


Fig. 1. Diagram of rules for FGA testing process and subject grouping.

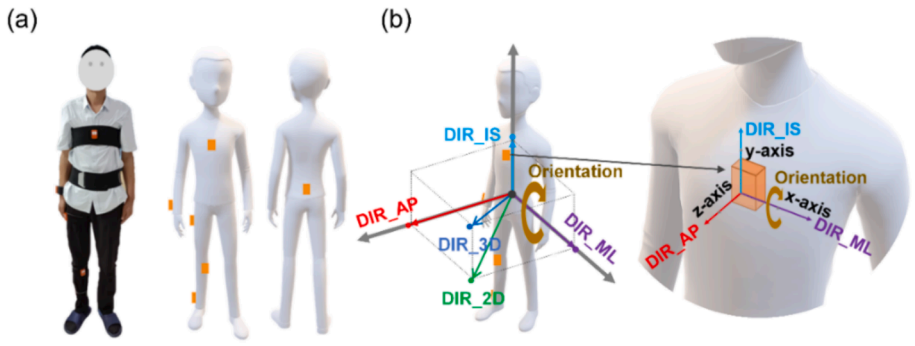


Fig. 2. Schematic diagram of the IMU configurations: (a) placement locations of the six IMU sensors; (b) schematic diagram of IMU-derived data across various directions and orientations.

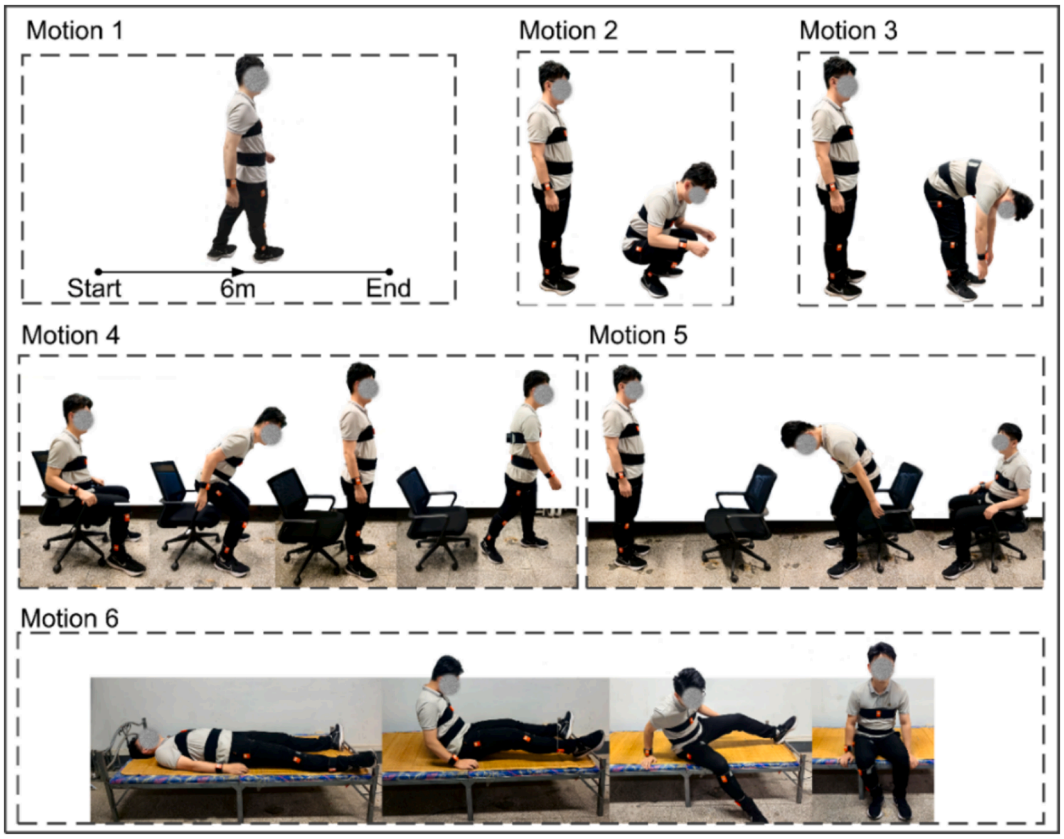


Fig. 3. Schematic diagram of experimental task B. Motion 1 (walking): Proceed straight for six meters at a regular walking pace; Motion 2 (squatting): Squat on heels with knees bent up close to the body; Motion 3 (bending): Bend over at the waist with upper limb sagging. Motion 4 (standing up): Rise from a chair and walk straight for three meters; Motion 5 (sitting down): Approach the chair and sit; Motion 6 (getting up): Rise from bed.

Table 1
Data labeling and criteria.

Label	Fall-risk	Criterion
Positive sample	High	FGA score $\leq$ 22
Negative sample	Low	FGA score $>$ 22

n sets of positive samples to form the training sample dataset; if $n > m$, randomly select m sets of samples from the positive samples, along with m sets of negative samples to form the training sample dataset.

2.2.2. Feature selection and extraction

Drawing from prior studies (Ghislieri et al., 2019; Diao et al., 2020), the subsequent thirteen features were chosen and derived from the raw acceleration data (see Table 2). To evaluate the performance of ML classifiers with various features selections and/or IMUs combinations, the thirteen features were divided into three sets, and the six sensors were organized into multiple combinations, as shown in Table 2.

2.2.3. Classifier hyperparameter optimization and performance evaluation

The classifiers used for model training included Support Vector Machine (SVM) (Kiprijanovska et al., 2020), Logistic Regression(LR) (Lemeshow & Hosmer, 1982), Random Forest (RF) (Breiman, 2001), Naive Bayes (NB) (Zhang, 2005) and K-nearest neighbor (KNN) (Gupta & Goel, 2020). The ML-based fall risk classification and all related data processing were performed using Python3.9 (Google Guido van Rossum). For each classifier, the grid search method was used for hyperparameter tuning, as shown in Table 3.

Table 2
Feature sets and IMU configuration for the training of the ML model.

Items	Abbreviation		Explanation
Feature selection	FE1		Combination of FE2 and FE3
	Time-domain and frequency-domain		
	FE2		Range
	Time-domain		Standard deviation
			Variance
			Mean
			Frame num
			Root mean squared
	FE3		First five FFT coefficients
	Frequency-domain		Skewness
			Kurtosis
IMU configuration	Single sensor	Sen 1	Right knee
		Sen 2	Right upper leg
		Sen 3	Right wrist
		Sen 4	Right ankle
		Sen 5	Sternum
		Sen 6	Pelvis
	Sensor combinations	Combinations of sensors are denoted by combining their numerical identifiers. For instance, a combining IMU configuration Sensor 1 and Sensor 2 is labeled as Sen 12	

Table 3
Hyper-parameter tuning methods for each classifier.

Classifier	Hyper-parameter	Tuning methods
SVM	kernel	rbf, poly
	cost	10
	gamma	0.001, 0.0001
	degree	1, 2, 3
LR	penalty	12
	class_weight	balanced
	solver	newton-cg, lbfgs, liblinear, sag
	cost	0.001, 0.01, 0.1, 1, 10, 100, 1000
RF	n_estimators	100 ~ 301, step width 10
	random_state	50 ~ 151, step width 10
KNN	n_neighbors	1 ~ 20, step width 1
	weights	uniform, distance
NB	kernel	Gaussian Naive Bayes, Bernoulli Naive Bayes, Multinomial Naive Bayes

To train the ML models and test their performance, the datasets were partitioned into random training sets and test sets. In this study, randomly selected four samples of data were allocated for testing, while the remaining twenty-four sample data were designated for training. A K-fold cross-validation (Stone, 1974), with k set to 5, was utilized to assess the ML models' performance. The dataset was divided into k subsets, with one subset acting as the validation set and the other k-1 subsets forming the training set. This procedure was carried out for every subset, totaling k iterations. Subsequently, the average error from all iterations was computed to determine the most suitable configuration.

Accuracy was used for quantifying the classification performance of the fall-risk assessment ML model, as defined by Eq. (3) (Attal et al., 2015).

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (3)$$

where *TP* and *TN* are the number of true positive and true negative, respectively, *FP* is the number of negative instances that were classified as positive, *FN* is the number of positive instances that were classified as negative.

2.3. Walking recognition based on deep learning

In previous studies (Senden et al., 2012; Jebelli et al., 2016) on fall risk assessment, walking gait stability has frequently been used to assess the fall risk of individuals. The results of this paper also indicate that the fall risk assessment model utilizing walking motion data outperforms those based on other types of motion (see 3. Result). This study aims to employ walking motion data for a further analysis of fall risk assessment. Consequently, there is a need to develop a walking recognition algorithm that can distinguish walking data from other motion data for accurate fall risk evaluation.

2.3.1. Architecture design of deep neural network

The deep learning model for walking recognition was developed based on studies by Yu et al. (2020). To minimize human decision-making and enhance accuracy, this study constructed DNN in two ways: (1) Inputting the original sensor data into the convolutional layer without feature processing; (2) Constructing multiple convolutional layers integrated with Long Short Term Memory (LSTM) (Greff et al., 2017) to learn deeper features and improve recognition accuracy. The Convolutional Neural Network (CNN) (Albawi et al., 2018) demonstrates exceptional proficiency in identifying local spatial coherence and dependencies in data. Concurrently, the introduction of scale invariance through max-pooling layers aids in mitigating overfitting, learning extra hidden features, and augmenting recognition accuracy.

The CNN-LSTM architecture is concisely depicted in Fig. 4. The raw data undergoes processing through two 1D convolutional layers, each containing 100 filters and 10 kernels. This is followed by a max-pooling layer featuring a pool size of 2, resulting in an output matrix size that is half of the input matrix. Subsequently, the convolved data is passed through two LSTM layers, with 300 and 200 units respectively, in order to capture long-term temporal dependencies within the time series. As LSTMs are prone to overfitting, a dropout layer with a 0.2 dropout

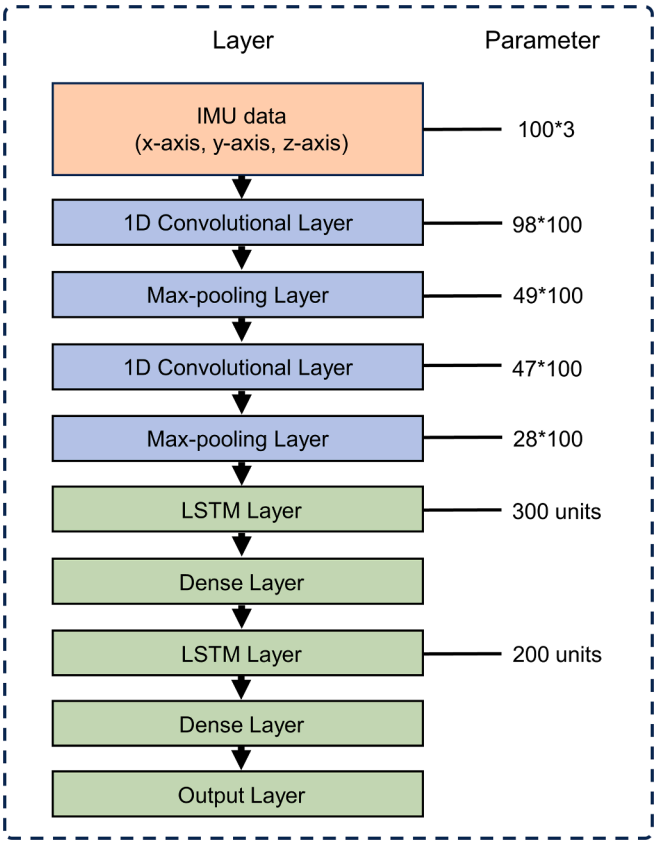


Fig. 4. Architecture and parameter of deep network CNN-LSTM.

probability is incorporated after each LSTM layer, signifying that 20 % of neurons are assigned a weight of zero. The network undergoes training for 50 epochs (allowing early termination) with a batch size of 100.

2.3.2. Data preprocessing and performance evaluation

Acceleration data gathered from 28 participants underwent filtering using a fourth-order zero-phase low pass Butterworth filter. The resulting filtered acceleration data in the DIR_3D direction, obtained from 6 independent IMUs, were subsequently used to train the neural networks. The walking motions were labeled as positive samples, and the other five motions were labeled as negative samples. For positive sample data sets, this study segmented the continuous walking motion of subjects into individual gait cycles as data samples to augment the sample size of the training dataset. The gait cycle was segmented by identifying data peaks (Pitt et al., 2020). Specifically, the orientation data parallel to the direction of travel, collected by the IMU at the right knee, was utilized for gait cycle segmentation (see Fig. 5).

In order to evaluate the performance of the CNN-LSTM, the dataset was divided into random training and testing sets, maintaining an 8:2 proportion, and 5-fold cross-validation was implemented to random training set. Early termination was enabled through callback monitors: halting the training process if the model failed to improve for two consecutive epochs, thereby preventing overfitting. Accuracy, precision, recall, specificity, and F1-score were used for computing the performance of walking recognition deep network CNN-LSTM, according to Eq. (3) to Eq. (7) (Attal et al., 2015). Precision evaluated the proportion of real positive samples (walking motion data) among all positive samples predicted. Recall assesses the number of true positive samples accurately predicted as positive (Powers, 2020). The F1-score representing the harmonic mean of precision and recall (Folorunso et al., 2022). This metric is useful in scenarios where balanced consideration of both false positives and false negatives is essential. F1-score was used as the main metric in this study to evaluate the performance of the deep neural network.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \tag{4}$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \tag{5}$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) \tag{6}$$

$$\text{F1-score} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall}) \tag{7}$$

where *TP* and *TN* are the number of true positive and true negative, respectively, *FP* is the number of negative instances that were classified as positive, *FN* is the number of positive instances that were classified as negative.

In summary, Fig. 6 shows the comprehensive process of the fall-risk assessment model proposed in this study: A. Acquisition of kinematic data using IMUs. B. Walking motion data recognition from kinematic data based on deep learning and single gait segmentation. C. Assessment of fall risk levels for walking motion data using ML.

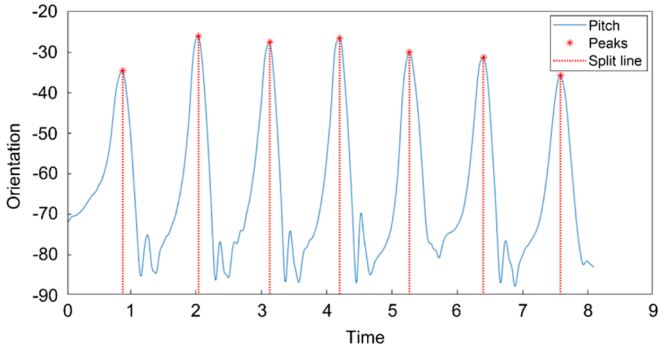


Fig. 5. Diagram of gait cycle division result. The data encompassed by two peak points (marked with *) represents a single gait cycle.

3. Results

3.1. Overview of sample data

Table 4 shows the detailed information and FGA scores of the twenty-eight subjects. There are eight positive samples (high fall-risk) and twenty negative samples (low fall-risk). Fig. 7 illustrates the sample data size of random training sets (including the training sets and validation sets) after undersampling, as well as the test sets for ML. 3318 groups of walking motion data and 480 groups of other motion data were used to train ML classifiers. 684 groups of sample data from two subjects with high-fall risk and 681 groups of sample data from two subjects with low fall-risk were used as test set which were only used for testing and were not used in undersampling or model training.

Fig. 8 shows the sample data size of the training and validation sets, and test sets. The random training sets include 4319 groups of positive samples and 3360 groups of negative samples which were used for DNN training. The test sets include 1080 groups of positive samples and 840 groups of negative samples, which were only used for testing the performance of DNN.

3.2. Fall-risk assessment based on ML

To develop the best performing fall-risk assessment model through ML, this study investigated the impact of motion type, feature selection, and IMU configuration on improving the fall-risk assessment accuracy of ML models.

3.2.1. Impact of motion type on fall-risk assessment performance

Table 5 shows the performance evaluation results of models separately trained using six types of motion datasets. The training sets were acceleration data in the DIR_3D processed by FE1 features. As shown in Table 5, for accuracy of classifiers trained with Motion 1, the maximum accuracy was 91.30 % (RF, Sen 2), while the minimum was 44.93 % (RF, Sen 5). For Motion 2, the maximum accuracy was only 80.00 % (SVM, Sen 4) and the minimum was 25.00 % (SVM, Sen 1 or Sen 2). For Motion 3, the maximum accuracy was 81.25 % (NB, Sen 5, or Sen 6), while the minimum was 31.25 % (KNN, Sen 2). For Motion 4, the maximum accuracy was 87.50 % (SVM, Sen 1, or RF, Sen 3, or KNN, Sen 1), and the minimum was 50.00 % (KNN, Sen 6). For Motion 5, the maximum accuracy was 87.50 % (KNN, Sen 2), and the minimum was 43.75 % (RF, Sen 6). For Motion 6, the maximum accuracy was 75.00 % (LR, Sen 3 or Sen 5), and the minimum was 43.75 % (RF, Sen 5). Among the six motion types, the maximum accuracy of assessment models trained with motion 1, motion 4, and motion 5 exceeds 85 %, while the minimum accuracy of assessment models trained with motion 2 and motion 3 falls below 40 %.

Upon analyzing the accuracies of classifiers using different motion data, this paper observed that classifiers based on data of motion 1 (walking), motion 4 (standing up), and motion 5 (sitting down) exhibit better classification performance. The classifier based on Motion 1 (i.e. Walking) has the highest overall accuracy (maximum 91.3 %). Moreover, both Motion 4 and Motion 5, which also exhibit a good performance, involve walking motion. Consequently, the subsequent study will focus on the performance of the fall risk assessment algorithm utilizing walking motion.

3.2.2. Impact of feature selection on fall-risk assessment performance

The fall-risk assessment model, based on the walking dataset, exhibited satisfactory performance. For the fall-risk assessment model trained on walking motion data, this paper examined the impact of feature selections on model performance.

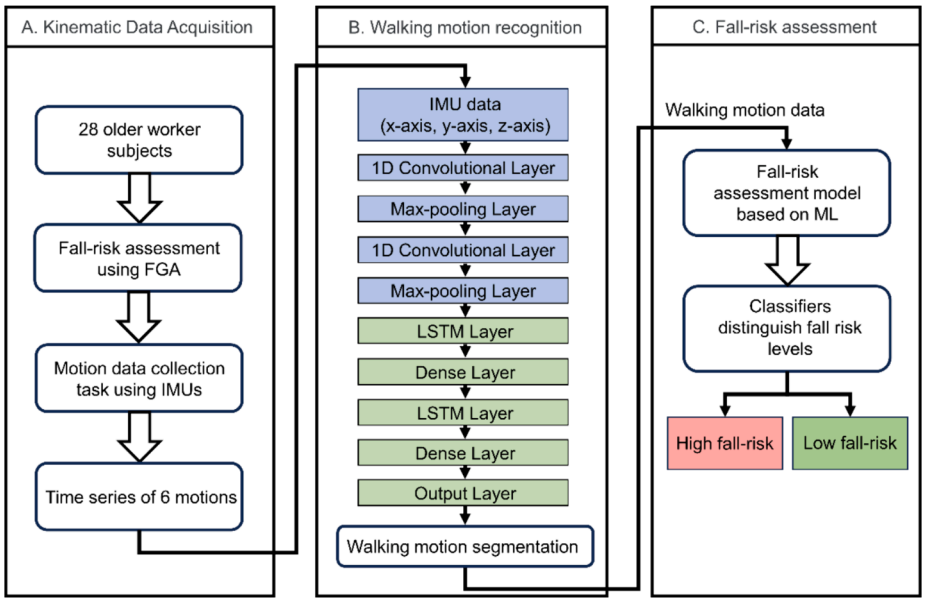


Fig. 6. Schematic representation of the fall-risk assessment model and the CNN-LSTM.

Table 4
Basic information of participants and FGA results.

No.	Age	FGA score	Label for ML	No.	Age	FGA score	Label for ML
1	62	19	Positive sample	15	76	17	Positive sample
2	60	26	Negative sample	16	67	26	Negative sample
3	65	25	Negative sample	17	70	23	Negative sample
4	60	26	Negative sample	18	80	16	Positive sample
5	60	26	Negative sample	19	68	19	Positive sample
6	68	25	Negative sample	20	70	23	Negative sample
7	74	24	Negative sample	21	77	15	Positive sample
8	65	27	Negative sample	22	77	11	Positive sample
9	67	26	Negative sample	23	75	23	Negative sample
10	73	25	Negative sample	24	76	14	Positive sample
11	64	25	Negative sample	25	70	25	Negative sample
12	64	26	Negative sample	26	69	23	Negative sample
13	71	26	Negative sample	27	74	17	Positive sample
14	65	27	Negative sample	28	70	28	Negative sample

Table 6 presents the accuracy of classifiers based on different feature selections. Classification accuracy across diverse feature sets and IMU configurations, utilizing various classifiers, ranged from 37.88 % to 91.30 %. The maximum accuracy was 91.30 % (FE1, Sen 2, and RF), while the minimum accuracy was 37.88 % (FE3, Sen 3, and SVM). The average classification accuracy shows no substantial difference between using FE1 and FE2 (with a difference of 1.14 %-9.08 % in average), but shows relatively larger difference between using FE1/FE2 and FE3(7.33 % to 27.82 %, excluding Sen 2). This indicates lower average classification accuracy with FE3. Nevertheless, the classification performance using Sen 2 based on FE3 was generally higher, with a maximum accuracy of 88.41 %.

3.2.3. *Impact of IMU configuration on fall-risk assessment performance*

The aforementioned analysis of Table 6 demonstrates that IMU configurations also influence the performance of fall risk assessment models. Table 6 shows that the accuracy of classifiers using Sen 2 and Sen 4 was significantly higher (exceeding 85.51 %) in terms of maximum classification accuracy for each feature set, compared to using Sen 1, Sen 3, Sen 5, and Sen 6 (below 78.79 %).

For two IMUs combinations, the result (see Table 7) indicates that the maximum accuracy was 88.41 % using Sen 26 (combination of Sen 2 and Sen 6), which was between the classification accuracy with Sen 2 only (91.3 %) and that with Sen 6 only (68.12 %). The accuracy for the IMU configuration Sen 24 (LR) was 88.40 %, which was higher than the accuracy of using Sen 2 alone (86.96 %) and Sen 4 alone (85.51 %), but also lower than the accuracy of using Sen 2 (RF) with 91.30 %. For three IMUs combinations, the maximum accuracy was 86.96 % with Sen 126 (combination of Sen 1, Sen 2, and Sen 6), which was higher than that with Sen 1 (68.12 %) or Sen 6 (68.12 %) only. But it was below than that with Sen 2 (91.30 %) only. In comparison to using single IMU configurations, utilizing combined IMU configurations did not yield significant improvements in the performance of fall-risk assessment model, and will increase the computational complexity.

3.3. *Walking recognition based on deep learning*

The prior analysis revealed that the fall-risk assessment model exhibited better performance using walking motion data. As walking motion recognition serves as the foundation for implementing fall-risk assessment model based on walking motion, this paper developed the walking recognition algorithm based on deep learning. Meanwhile, for the sake of portability, our research explored the capability of one single IMU to recognize walking motion. Table 8 displays the test performance of the CNN-LSTM model using motion data collected independently by six IMU sensors. Generally, the evaluation metrics indicated high performance of the CNN-LSTM model for recognizing walking motion using acceleration data. The F1-score and accuracy for all IMU configurations exceeded 90 % and in terms of all evaluation metrics, both Sen 2 and Sen 6 configurations surpassed 90 %. The maximum F1-score, 95.18 %, was achieved using Sen 6. The minimum F1-score, 91.20 %, was using Sen 4.

Fig. 9 presents the confusion matrix for the walking recognition task, with the FN values (86 to 165) being slightly higher than the FP values (20 to 46). Further analysis revealed that, although all models

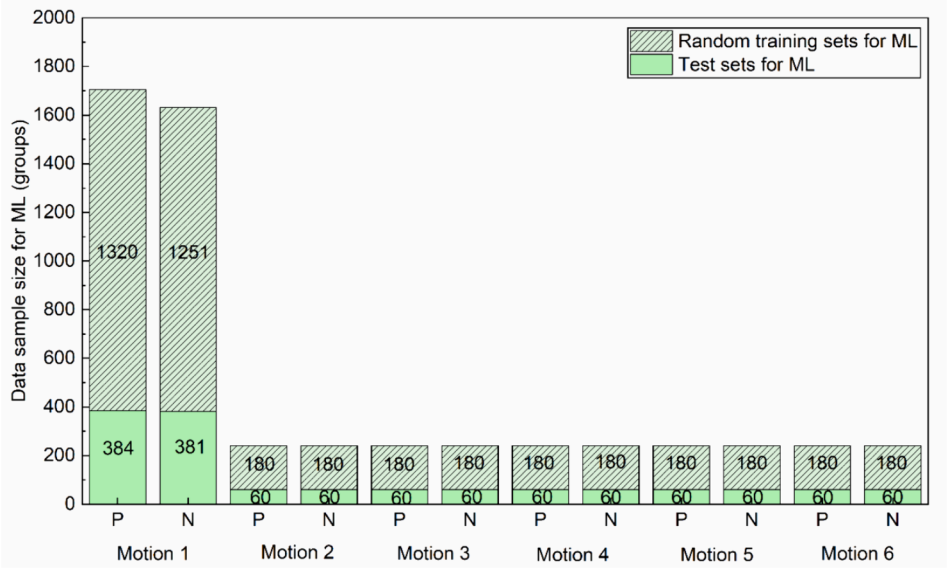


Fig. 7. Dataset sample size for ML. P: positive sample, N: negative sample.

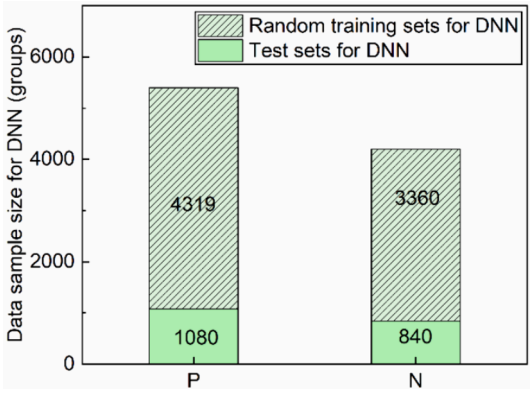


Fig. 8. Dataset sample size for DNN. P: positive sample, N: negative sample.

demonstrated commendable overall performance in the recognition of walking data, Sen 1, Sen 3, Sen 4, and Sen 5 exhibited a higher degree of confusion in comparison with Sen 2 and Sen 6 in the identification of false negative samples (FN all exceeding 100), which means many positive samples are incorrectly predicted as negative samples. Notably, Sen 6 displayed the most optimal performance in this aspect, having only 88 false negative samples (FN = 88) and 20 false positive samples (FP = 20), thereby showing a relatively lower level of confusion.

4. Discussion

Falls significantly contribute to both fatal and nonfatal injuries in aged workers. Assessing the fall risk for aged workers is crucial in enhancing occupational safety and public well-being (Varianou-Mikellidou et al., 2019). In this paper, we gathered IMU-based kinematic data of aged workers with different fall risk levels through experimental means, aimed to develop fall-risk assessment models based on ML and IMU for aged workers. The result indicates that the developed IMU-based ML method exhibits excellent performance in assessing fall risk, potentially allowing individuals to assess their own fall risk before clinical diagnosis by embedding it within a single wearable IMU device. The maximum accuracy reached 91.3 % with an optimized RF classifier. The performance of fall-risk assessment models varies with different training datasets (IMU configurations, feature selections), as well as the classifiers used.

In the study on using IMU sensors for fall risk assessment in older adults, finding the optimal combination of sensor placement, test task, and feature selections are key to creating effective methods (Montesinos et al., 2018). Therefore, this study examined the differences in the performance of fall-risk assessment models using different motion types of data, IMU configurations, classifiers, and training data feature selection. The purpose is to identify the optimal assessment model, considering assessment performance, IMUs wearing convenience, data usage, and computational complexity. For the motion type, as illustrated in Table 5, the performance of the trained assessment model varies significantly when using different types of motion data, despite employing the same classifier and IMU configuration. For instance, the assessment model using motion 2 data (with Sen 1, SVM, FE1) exhibits the minimum accuracy at 25.00 %. However, the accuracy improved to 87.50 % when using motion 4 data (with Sen 1, SVM, FE1). The results show that the accuracy based on data from motion 1, motion 4, and motion 5 was relatively higher than that based on the other three motions data. Motions 1, 4, and 5 all incorporated normal walking, whereas the other three motions did not. This result suggests that walking motion data might be more suitable for developing a fall-risk assessment model (Promsri et al., 2023). As stated in previous studies (Robinovitch et al., 2013; Nevitt and Cummings, 1993), instability during walking is a primary cause of actual falls, community-dwelling seniors were more likely to fall during walking. Walking motion data can be more readily obtained and identified in daily life and occupational settings (Yang & Hsu, 2010). Several studies (Bautmans et al., 2011; Senden et al., 2012) have demonstrated the potential of walking data in predicting fall risk by identifying gait-related risk factors. Therefore, it is recommended to use walking motion acceleration data for fall risk assessment model development.

The reason why walking motion data works better for developing fall-risk assessment models might be attributed to the base of support is narrowest in the DIR_ML direction when walking (Brodie et al., 2015), compared to the other motion types. Thus, the high fall-risk groups might exhibit more pronounced change of acceleration amplitude in the mediolateral direction (DIR_ML) while walking forward. In addition, several researches (Menz et al., 2003; Yack and Berger, 1993) have also reported on the efficacy of anteroposterior direction (DIR_AP) and/or DIR_ML acceleration-based parameters in gait instability. This study would like to investigate the most effective acceleration direction, including 2D and 3D synthetic directions, for developing optimal fall-risk assessment models. Thus, this study further compared the

Table 5
Classification accuracy of different classifiers based on acceleration data from different motions in DIR_3D direction.

Motion type	IMU Configuration	Classifier				
		SVM	LR	RF	KNN	NB
Motion 1	Sen 1	72.73 %	71.01 %	68.12 %	63.77 %	57.97 %
	Sen 2	78.79 %	86.96 %	**91.30 %**	69.57 %	68.12 %
	Sen 3	72.73 %	73.91 %	75.36 %	60.87 %	69.57 %
	Sen 4	86.36 %	85.51 %	73.91 %	76.81 %	75.36 %
	Sen 5	59.09 %	62.32 %	**44.93 %**	71.01 %	53.62 %
	Sen 6	78.79 %	68.11 %	68.12 %	68.12 %	55.07 %
Motion 2	Sen 1	**25.00 %**	31.25 %	50.00 %	31.25 %	50.00 %
	Sen 2	**25.00 %**	31.25 %	37.50 %	31.25 %	50.00 %
	Sen 3	50.00 %	62.50 %	62.50 %	50.00 %	56.25 %
	Sen 4	**80.00 %**	50.00 %	50.00 %	60.00 %	10.00 %
	Sen 5	75.00 %	62.50 %	56.25 %	50.00 %	56.25 %
	Sen 6	43.75 %	62.50 %	62.50 %	43.75 %	50.00 %
Motion 3	Sen 1	43.75 %	56.25 %	50.00 %	37.50 %	43.75 %
	Sen 2	37.50 %	43.75 %	50.00 %	**31.25 %**	43.75 %
	Sen 3	43.75 %	62.50 %	68.75 %	68.75 %	68.75 %
	Sen 4	70.00 %	80.00 %	50.00 %	50.00 %	50.00 %
	Sen 5	80.00 %	80.00 %	68.75 %	68.75 %	**81.25 %**
	Sen 6	68.75 %	75.00 %	68.75 %	43.75 %	**81.25 %**
Motion 4	Sen 1	**87.50 %**	68.75 %	68.75 %	**87.50 %**	81.25 %
	Sen 2	62.50 %	62.50 %	68.75 %	75.00 %	62.50 %
	Sen 3	81.25 %	81.25 %	**87.50 %**	81.25 %	81.25 %
	Sen 4	71.42 %	60.71 %	67.86 %	85.71 %	85.71 %
	Sen 5	75.00 %	75.00 %	71.43 %	68.75 %	62.50 %
	Sen 6	62.50 %	68.75 %	75.00 %	**50.00 %**	68.75 %
Motion 5	Sen 1	81.25 %	81.25 %	81.25 %	62.50 %	75.00 %
	Sen 2	75.00 %	75.00 %	62.50 %	**87.50 %**	75.00 %
	Sen 3	56.25 %	50.00 %	75.00 %	50.00 %	50.00 %
	Sen 4	71.43 %	60.71 %	75.00 %	82.43 %	85.71 %
	Sen 5	56.25 %	56.25 %	56.25 %	56.25 %	62.50 %
	Sen 6	56.25 %	62.50 %	**43.75 %**	62.50 %	56.25 %
Motion 6	Sen 1	62.50 %	68.75 %	56.25 %	62.50 %	62.50 %
	Sen 2	62.50 %	62.50 %	62.50 %	56.25 %	56.25 %
	Sen 3	68.75 %	**75.00 %**	56.25 %	62.50 %	62.50 %
	Sen 4	60.71 %	57.14 %	50.00 %	60.71 %	57.14 %
	Sen 5	68.75 %	**75.00 %**	**43.75 %**	62.50 %	62.50 %
	Sen 6	56.25 %	62.50 %	62.50 %	68.75 %	50.00 %

performance of fall-risk assessment models trained by acceleration data in different directions. We incorporated DIR_AP, DIR_ML, DIR_2D and DIR_3D data (see Fig. 2(b)) into the test. The ML process was performed again, using data in various directions. Fig. 10 shows the results of accuracy of models trained by different direction data. The upper whisker and upper quartile show minimal differences in accuracy between assessment models with the DIR_AP datasets and DIR_ML datasets (difference was −1.45 %, 0.10 %). All these quartiles of accuracy for assessment models with the DIR_3D datasets were the highest. It may because that the synthetic direction data, particularly the DIR_3D direction dataset, contains more signal characteristics related to physical instability, making it more suitable for developing fall-risk assessment models. This study developed the fall-risk assessment models using the DIR_3D direction data achieving a maximum accuracy of 91.30 %, better than that developed by Howcroft et al. (2017) used DIR_AP and DIR_ML direction datasets (achieving a maximum accuracy of 78.90 %.). Therefore, it is recommended to use DIR_3D directions to conduct related IMU-based fall risk assessment research.

For feature selection of training dataset, this study examined the effect of feature selections of the time-domain features (FE2), frequency-domain features (FE3), and their combination (FE1) on the performance of the fall-risk assessment. Overall, the classification performance employing FE1 and FE2 was significantly better. The accuracy using only the FE3 was generally poorer in overall. However, under certain configuration combinations, the accuracy using FE3 surpassed that using FE2, for instance, the combination of FE3-Sen 2-NB achieved 88.41 % accuracy, which is higher than FE2-Sen 2-NB at 68.12 %. This indicates that, while time-domain features typically offer better accuracy, a comprehensive analysis considering the combination with IMU configurations and classifiers is also essential in practical applications. The review study by Montesinos et al. (2018) also confirmed that the effect of feature selection, especially frequency-domain features, for assessing gait stability will be affected by the placement of the sensors. Moreover, the combination of time-domain and frequency-domain feature set (FE1) exhibited slightly higher accuracy compared to using only the time-domain feature set (FE2). Nevertheless, certain combinations rendered the accuracy using FE2 higher than that of FE1, such as the combination of FE2-Sen 1, which slightly outperformed the FE3-Sen 1. In feature selection for IMU-based fall-risk assessment tasks, a comprehensive consideration of the IMU configuration and classifiers is advised. This study recommends the following configuration combinations (task-feature selection-IMU configuration-classifier) for fall risk assessment of aged workers: Walking-FE1-Sen 2-RF; Walking-FE2-Sen 2-LR; Walking-FE3-Sen 2-NB.

This paper also examined the effect of various IMU configurations on the performance of fall-risk assessment ML models. The IMUs in this paper were attached to lower limbs (Sen 1, Sen 2, Sen 4), upper limbs (Sen 3), and trunk (Sen 5, Sen 6). The results indicate that model accuracy under lower limbs configurations was higher compared to those under upper limb or trunk configurations. Our results are consistent with the study of Montesinos et al. (2018), whose walking test results demonstrated that the most compelling sensor placement to assess fall risk was lower limbs. In contrast, for the quiet standing posture stability test, the pelvis was the most effective sensor placement location to assess fall risk. Because the pelvis is near the center of mass, corresponding to COP-based measures, the golden standard for postural stability (Ghislieri et al., 2019), contains more data variation features (Ishigaki et al., 2011). Drawing an analogy from the walking tests to above analysis, for dynamic motion, our results suggest that the performance of classifiers using pelvis IMU configuration was poorer than that using lower limber (upper leg and ankle) data. This finding may imply the IMU-based motion data of lower limbs have more potential features for using ML methods to assess fall risk during walking. In a multitude of IMU-based fall-risk assessment and gait stability studies (Caby et al., 2011; Greene et al., 2010; Doheny et al., 2011), scholars have demonstrated that the fall risk and gait stability can be effectively distinguished by analyzing

Table 6
Classification accuracy for various feature selections using different classifiers based on walking motion acceleration data in DIR_3D direction.

Feature Set	IMU Configuration	Classifier					
		SVM	LR	RF	KNN	NB	Average
FE1	Sen 1	72.73 %	71.01 %	68.12 %	63.77 %	57.97 %	66.72 %
	Sen 2	78.79 %	86.96 %	**91.30 %**	69.57 %	68.12 %	78.95 %
	Sen 3	72.73 %	73.91 %	75.36 %	60.87 %	69.57 %	70.49 %
	Sen 4	86.36 %	85.51 %	73.91 %	76.81 %	75.36 %	79.59 %
	Sen 5	59.09 %	62.32 %	**44.93 %**	71.01 %	53.62 %	58.19 %
	Sen 6	78.79 %	68.11 %	68.12 %	68.12 %	55.07 %	67.64 %
FE2	Sen 1	77.27 %	75.36 %	69.57 %	63.77 %	59.42 %	69.08 %
	Sen 2	60.61 %	**86.96 %**	84.06 %	59.42 %	56.52 %	69.51 %
	Sen 3	71.21 %	71.01 %	69.57 %	72.46 %	73.91 %	71.63 %
	Sen 4	71.21 %	85.51 %	76.81 %	71.01 %	69.57 %	74.82 %
	Sen 5	50.00 %	63.77 %	**46.38 %**	55.07 %	59.42 %	54.93 %
	Sen 6	74.24 %	69.57 %	68.12 %	60.87 %	59.42 %	66.44 %
FE3	Sen 1	50.00 %	53.62 %	52.17 %	50.72 %	55.07 %	52.32 %
	Sen 2	74.24 %	76.81 %	85.51 %	75.36 %	**88.41 %**	80.07 %
	Sen 3	**37.88 %**	40.58 %	62.32 %	39.13 %	39.13 %	43.81 %
	Sen 4	60.61 %	62.32 %	60.87 %	50.72 %	71.01 %	61.11 %
	Sen 5	40.91 %	50.72 %	39.13 %	57.97 %	49.28 %	47.60 %
	Sen 6	50.00 %	47.83 %	44.93 %	46.38 %	46.38 %	47.10 %

Table 7
Classification accuracy for various multiple IMU configuration using different classifiers based on walking motion acceleration data with FE1 in DIR_3D direction.

IMU Configuration	Classifier					
	SVM	LR	RF	KNN	NB	Average
Sen 12	75.76 %	76.81 %	84.05 %	65.22 %	57.97 %	71.96 %
Sen 16	72.72 %	65.21 %	69.57 %	57.97 %	53.62 %	63.82 %
Sen 24	77.27 %	88.40 %	84.06 %	71.01 %	73.91 %	78.93 %
Sen 26	77.27 %	81.15 %	**88.41 %**	60.87 %	55.07 %	72.55 %
Sen 46	87.88 %	85.50 %	78.26 %	75.36 %	63.77 %	78.15 %
Sen 124	77.27 %	78.26 %	81.16 %	66.67 %	68.12 %	74.30 %
Sen 126	78.79 %	68.11 %	**86.96 %**	60.87 %	56.52 %	70.25 %
Sen 146	75.76 %	72.46 %	79.71 %	62.32 %	60.87 %	70.22 %
Sen 246	78.79 %	84.05 %	81.16 %	66.67 %	57.97 %	73.73 %
Sen 1246	78.79 %	71.01 %	81.16 %	65.22 %	59.42 %	71.12 %

Table 8
Performance of walking recognition models based on the data set of six IMU configurations.

IMU Configuration	Accuracy	F1-score	Precision	Recall	Specificity
Sen 1	93.59 %	93.96 %	97.66 %	90.53 %	89.33 %
Sen 2	94.09 %	94.53 %	96.63 %	**92.51 %**	91.24 %
Sen 3	92.48 %	92.98 %	95.76 %	90.35 %	88.93 %
Sen 4	90.87 %	91.20 %	97.44 %	85.71 %	84.68 %
Sen 5	92.49 %	92.87 %	97.52 %	88.64 %	87.44 %
Sen 6	**94.84 %**	**95.18 %**	**98.16 %**	92.38 %	**91.25 %**

and processing data from lower limb IMUs. Fewer studies placed IMU on the wrist to assess fall risk, they may apply to specific tasks. The upper limb and hand undertake many other irrelevant tasks in daily activities (Choi, 2022), which greatly interferes with the data of IMU on wrist. Unless the subjects are required to move their upper limbs according to regulations, which is unreliable in actual assessment. Table 6 shows that the performance of the fall-risk assessment models, trained based on wrist (Sen 3) IMU data, is also subpar (the maximum accuracy is only 75.36 %, based on FE1), generally less than models developed using lower limb (Sen 2 and Sen 4) IMU data. These results recommend lower limbs, especially upper leg and ankle, as the preferred IMU configuration locations for the application of ML methods in IMU-based fall-risk assessment. In summary, this study investigated the impact of motion data types of training data set, feature selections, and IMU configurations on the performance of ML models, with identified the optimal combination. Subsequently, a random forest classifier was trained and achieved an accuracy of 91.30 % after hyperparameter optimization. This result surpassed the accuracy of the assessment model developed by researchers using neural network algorithms (accuracy = 86.00 %) (Meyer et al., 2021) in the area of fall risk assessment research. Consequently, this study introduces a novel convenient and technology-driven solution for accurate fall risk assessment in the aged workers.

Furthermore, given the advantages of walking motion data in developing ML models for fall-risk assessment, this paper developed a deep learning-based walking motion recognition algorithm, to extract walking motion data as valid input data for fall-risk assessment classifiers from time series data of IMU. The parameters adjusted CNN-LSTM deep network (Jang et al., 2018) was used to distinguish walking motion data from the samples. The performance of CNN-LSTM in motion multi-classification has been proved by numerous studies (Zhang et al., 2022; Zhao et al., 2019; Yu et al., 2020), In this study, it was tentatively applied to binary classification, where the positive samples were walking motion data and negative samples were the other motions data. The results indicate that the algorithm could effectively extract walking motion data only using a single IMU (achieving a maximum F1-score of 95.18 %). The F1-value of the CNN-LSTM algorithm using Sen 6 datasets reached 95.18 %. This high F1-value indicates that the performance of the developed walking motion recognition algorithm was acceptable and more true sample datasets can be obtained with fewer errors. In addition, the F1-score of walking recognition algorithm using Sen 2 datasets reached 94.53 %, ranking second to Sen 6 datasets. Given that Sen 2 exhibits the best performance in fall-risk assessment, it also excelled in walking motion recognition. Therefore, the right upper leg with Sen 2 was recommended as the optimal body location with IMU configuration for IMU-based fall-risk assessment research.

The fall-risk assessment models developed in this study enables aged workers to determine their fall risk at any time without the need for specific tests, by simply extracting motion data from their daily behavior. Aged workers could use this IMU-based method to self-test their fall risk, screen out high-risk individuals cost-effectively, and implement timely interventions to avoid fall-related injuries. In further

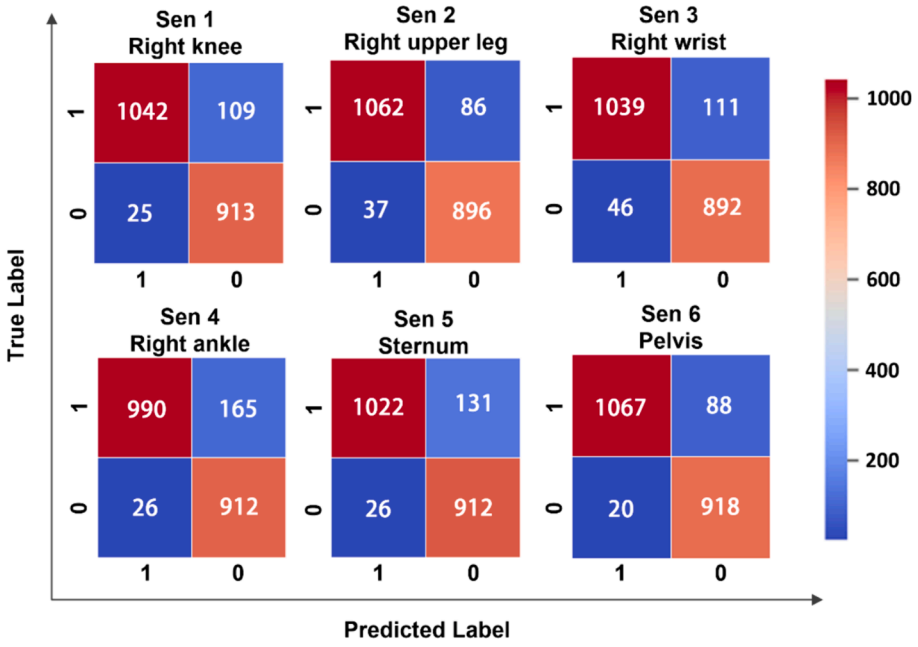


Fig. 9. Confusion matrix for the walking recognition task.

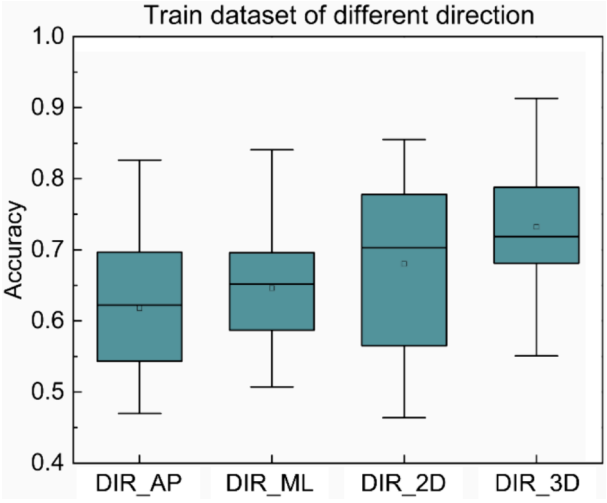


Fig. 10. Boxplot for accuracy of fall-risk assessment models trained by acceleration data in different directions. The dataset feature selection was FE1, IMU configuration includes Sen 1, Sen 2, Sen 4, and Sen 6.

research, the fall-risk assessment models should be integrated with hardware to explore its age-friendly design and usability. Furthermore, preventive measures should be provided for aged workers based on the results of fall risk assessment in the research. and enable real-time fall risk status accessed remotely by external parties, such as family members or employers.

Acknowledging the limitations of the research is necessary. Firstly, this study aimed to compare the impact of different feature selections in the time-domain and frequency-domain on the development of fall risk assessment ML models, so feature engineering was not carried out in depth. Simply grid search method was used in the hyper-parameter tuning of ML, which is just an attempt to apply the ML method to the IMU-based fall-risk assessment. Future research could explore more advanced methods to enhance the accuracy of these classifiers. Secondly, this study employed only one experienced professional to evaluate the FGA test, which may introduce personal bias in the evaluation results, potentially affecting the accuracy of assessment models. Small sample size is also one of the limitations that may affect the generalizability of the results. In subsequent research, a long-term follow-up survey spanning several months could be conducted with more aged workers to validate the reliability of fall-risk assessment models and to further optimize models.

5. Conclusion

In order to develop practical, technology-driven solutions for the pre-identification of fall risks among the aged workers, this study explored a method for assessing fall risk by integrating ML techniques with wearable IMUs. Our results revealed the significant potential of using a single IMU combined with ML to accurately assess fall risks, achieving an accuracy of 91.30 %. This result validates the effectiveness of our method. Notably, the optimized RF classifier and CNN-LSTM model demonstrated the potential of embedding such technologies in wearable devices, aiding in self-assessment of fall risks and timely interventions. The use of the walking motion data collected by the IMU attached to the right upper leg, incorporating features from both time and frequency domains, was also recommended for research concerning IMU-based fall-risk assessment. Our approach has proven the feasibility of using machine learning and wearable technology to assess the fall risks of the aged workers. Future research should further explore the age-friendly design of assessment algorithms to make artificial intelligence technology truly accessible to the aged workers, allowing for timely interventions to prevent fall-related injuries and enhance production safety for the aged workers.

CRediT authorship contribution statement

Junhui Kou: Writing – original draft, Methodology, Investigation. **Xin Xu:** Software, Investigation, Formal analysis. **Xiaoyang Ni:** Supervision, Resources. **Shengkai Ma:** Visualization, Software. **Liangjie Guo:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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